Why Simple Laws Are Most Probable: Occam's Razor, All Is Math, and Discrete Time Steps Forward

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Abstract

Simple laws are more probable because every bit of a complete description is a yes-or-no that no rule decides, so each added bit halves the description's probability. A shortest description has no pattern left to compress, so its bits are random in the sense of the rule-random dichotomy: nothing favors 0 over 1, and no bit depends on another, since a preference or a dependence would itself be a rule that has to be written down. Because every structure exists, the half is a count and not a guess: among all strings, one in 2^k begins with any given k bits. Longer descriptions are more numerous and still lose in total, because whether to add another bit is a second unruled yes-or-no, so everything longer than a given length weighs exactly as much as that length: the tail equals the peak. Added laws more often break the conditions observers need than create them, so at least half the weight sits at the shortest length that can produce observers. The universe therefore starts simple, laws and starting state together, and since minds like ours are complex, it has to grow them. Growth takes discrete causal steps, so time runs forward from that start.

Keywords: Simplicity Dominance Principle, algorithmic probability, Kolmogorov complexity, Occam's Razor, principle of indifference, measure problem, rule-random dichotomy, mathematical monism

1. The Claim

Simpler laws are more probable, and this paper derives by how much: each bit of a complete description halves its probability. Call this the Simplicity Dominance Principle. Ray Solomonoff introduced the same weights as a starting point for predicting data (Solomonoff, 1964). Here they are derived rather than chosen, and they weigh what exists, not only what to expect next. Three results follow. Each bit halves the weight. Whether to add another bit is a second 50/50, so longer descriptions lose in total. At least half the weight sits at the shortest length that can produce observers, so the universe starts simple and has to grow the complex minds that observe it. It grows them in causal steps, a cause and then its effect, so time runs forward from the simple start (Section 6). The order of cause and effect is a priori; the simple start comes from the weighting, an argument with premises of its own.

The derivation starts from the rule-random dichotomy: everything that exists is ruled or random, and both are mathematical. A ruled structure is produced by a rule shorter than itself; a random one has no shorter rule. Anything outside both would be outside mathematics, and so outside logic, since nothing could be reasoned about it. Matheism adds that every consistent structure, one free of contradiction, exists (compare Schmidhuber, 2000; Tegmark, 2014). The question is then which structures an observer should expect to be in, and without a weight on structures nothing can be predicted at all.

The weight attaches to a complete description: the laws, the starting state and the rule connecting states to what observers record, with information they share counted once. A short equation whose starting state hides the real information is a long description, so moving complexity from the laws into the start saves nothing.

2. Why Each Bit Halves

Each bit of a shortest description is a yes-or-no that no rule decides, so 0 and 1 are equally likely. Stated precisely, a shortest description cannot be compressed: if it could, the compressed version plus an instruction to unpack it would be a shorter description of the same thing. Its bits are therefore random in exactly the sense of the dichotomy. With no rule to favor either value, each has probability 1/2. A lean toward 0 or 1 would itself be a rule, and would have to be written into the description, making it longer. A biased coin is mathematical too, but its bias is a number that has to be written down; one half is the only chance that needs no such number, because it is the only one that treats 0 and 1 alike.

No bit depends on another, for the same reason. Two bits can each be fair and still agree every time, as when one choice between all zeros and all ones fixes every bit at once. That agreement is a rule, "repeat the first bit," and a rule costs bits. With nothing written down, each bit is independent of the others.

Because every structure exists, the half is also a count and not a guess. Take all strings (sequences of 0s and 1s) of some length n. Exactly half begin with 0, and when n is at least k, exactly one in 2^k begins with any given k bits. The weight of a k-bit description is that share: the fraction of all strings that begin with it. On a computer the share is one of tapes: fill a universal machine's tape with fair coin flips, and a k-bit program begins one tape in 2^k. The machine reads no further than the program's end, so every tape that begins with it computes the same history. This is probability as Pascal and Fermat counted it in 1654, a share of equally possible cases (Hacking, 1975), and the count carries independence with it: a quarter of all strings begin with 0 and then 1, the product of the two halves, with no further assumption.

A particular description of L bits therefore has weight 2^(-L), and two complete accounts that differ by ΔL bits have prior odds (the odds before any evidence) of 2^(ΔL). Ten bits give 1,024. The evidence then multiplies each weight by how well

the account predicts what is observed. An account 20 bits shorter starts about a million times ahead and loses that lead if it makes the observations a million times less likely, so fit and length trade at one bit for each factor of two, and among accounts that fit equally well, the shortest dominates.

Random, in the dichotomy, means a structure with no shorter rule, a pattern nothing compresses. It does not mean an event without a cause. The 50/50 is a share of structures, not a coin flipped by nothing.

3. Why Longer Descriptions Lose in Total

Longer descriptions are more numerous, and they still lose in total, because whether to add another bit at all is a second unruled yes-or-no. If number decided, we should expect to find ourselves under an extremely long description, a world of endless special cases and arbitrary detail; we find ourselves under laws that fit on a page. Stated precisely, there are 2^n strings of length n, so if each counted as a complete description weighted 2^(-n), every length would carry a total weight of one and the sum over lengths would be infinite. Half is the gentlest rate any weighting can have: with equally long descriptions weighted alike, a factor milder than one half per bit would give each length more total weight than the one before, without limit.

The choice to stop or go on settles the total. It is unruled, so it is 50/50: half the weight stays with a description as it is, and half goes on to longer ones. All descriptions of n bits together then weigh 2^(-(n+1)), half as much as those one bit shorter, and everything longer than n bits, added up, weighs exactly as much as length n alone: the tail equals the peak.

Picture each universe as its own run of fair coin flips, a new experiment each time: one flip decides whether its description goes on, and the next decides the bit added. Any laws can come up in a run, yet half of the runs still going stop at each

step, so most runs end with short laws. A catalog that listed each set of laws once would favor long laws by sheer number. The runs favor short ones, because a long law has to win every one of its stop-or-go flips.

In this picture one particular description loses three quarters of its weight with each added bit, a half for going on and a half for the bit's value. A code that states the length once, in a short header, loses only about half per bit (Li and Vitanyi, 2019), so odds between particular descriptions in this paper use that smaller factor, while the totals by length use the stop-or-go weights. Either way, descriptions none of which begins another claim separate shares of all strings, so their weights add up to at most one.

4. At Least Half at the Shortest Length That Works

Added laws more often break the conditions observers need than create them, so at least half the weight sits at the shortest length that can produce observers. Producing observers means producing structures that record and model their surroundings. The test mentions none of our own laws, so the count does not build in its answer.

Stated precisely, under the stop-or-go weights of Section 3, let M be the shortest length that produces observers, and let s_n be the share of length-n descriptions that produce them. If no longer length has a larger share than s_M, length M carries at least half of the observer-producing weight, because the tail beyond M weighs exactly as much as length M and its share is nowhere larger. If the share also does not rise from one length to the next, lengths M and M+1 carry at least three quarters, and lengths M through M+2 at least seven eighths. Without that second condition, shares of one at M, none at M+1 and one after that leave the first two lengths only two thirds.

The condition does real work. Under the same weights, if one description in sixteen of length four produced observers and all thirty-two of length five did, length five would carry eight times the observer weight of length four. Physics points the other way. Small changes to the constants of our physics often remove stable nuclei, long-lived stars or chemistry (Barnes, 2012), and some lengths of description cannot produce observers at all; each such gap moves more weight onto the shortest length that works. For a single output, the coding theorem makes the same point: all its longer descriptions together cannot outweigh its shortest one by more than a fixed factor (Levin, 1974). The prediction is that the complete laws lie near the shortest description that can produce observers.

5. Objections

"The half is the principle of indifference, which leads to paradox." Joseph Bertrand's paradoxes arise for continuous quantities, which can be spread evenly in many ways, each a different measure (Bertrand, 1889). A bit has two values, and only one split treats them alike. Nor is the half a confession of ignorance: no rule exists to favor either value, and the count of Section 2 makes it a share of what exists.

"Incompressibility describes one string, not how strings are weighted." The weight does not come from one string alone. It comes from what exists: every string exists, and the share of them that begins with a given description is the count of Section 2. Incompressibility says why nothing inside the description tilts that share.

"A biased chance is just as mathematical." It is, and its bias is a number the description must carry. A description that carries it is longer, and its added bits face the same halving.

"The weights depend on the programming language." Universal languages, those able to express any computation, agree on lengths up to a constant that depends on the pair of languages and not on what is described (Kolmogorov, 1965), so the weights under two languages differ by at most a fixed factor, and each bit of real difference between two accounts still halves the weight. The 50/50 does not depend on the language; only the lengths do. Close comparisons state their language.

"Shortest lengths cannot be computed." In general they cannot, so comparisons use the lengths of descriptions actually written, which bound the shortest from above. The measure says what to compute, as a definition of probability does without listing every case.

"Why should existence come with weights at all?" An observer asking which structure it is in has no other way to answer. Every structure exists, so the question is what share of what exists is like this, and the count is the only answer that needs no further rule.

6. What Follows

The universe starts simple, laws and starting state together, because every bit of its description halves its weight. Minds like ours are complex, so their complexity has to grow from that start: each bit of a brain's organization is either written into the description, halving the weight, or grown by the laws from a simpler state, and the probable universes grow it. Many simple rules do. The Game of Life's one rule about neighbors grows five live cells, the R-pentomino, through more than a thousand generations, throwing off gliders (small patterns that travel while keeping their shape), and suitable patterns in it can run any computation (Gardner, 1970; Berlekamp, Conway and Guy, 1982). What grows is organization; the description stays short, since every later state is fixed by the

rule, the start and the number of steps. Growth takes causal steps, a cause and then its effect, since a cause produces its effect as the next state and a continuum has none. That is an a priori proof in the logician's sense. Russell held that motion is only being at different places at different times (Russell, 1903); a position at every instant with nothing producing the next is the emperor's new points. So time runs from the simple start toward the minds it grows. Each step is a rule applied to rules, and time is that application. That chain needs the simple start only as a fact, and observation shows it: the laws we know fit on a page, and the early universe was smooth to about one part in a hundred thousand, with no structures yet, let alone minds (Planck Collaboration, 2020). It stands even for a reader this paper does not persuade, since a reason can fail without taking with it the fact it explains.

Conclusion

Simple laws are more probable because everything that exists is ruled or random, so each bit of a complete description that no rule decides is a 50/50 choice, and since every structure exists, the half is a count of what exists. Each added bit halves a description's weight, the choice to stop or go on makes longer descriptions lose in total as well as one by one, and at least half the weight sits at the shortest length that can produce observers. The weights Solomonoff chose as a starting point for prediction follow from the dichotomy, and they weigh what exists. The universe starts simple and has to grow the minds that observe it.

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